Introduction and Basics

Python course - Class 1

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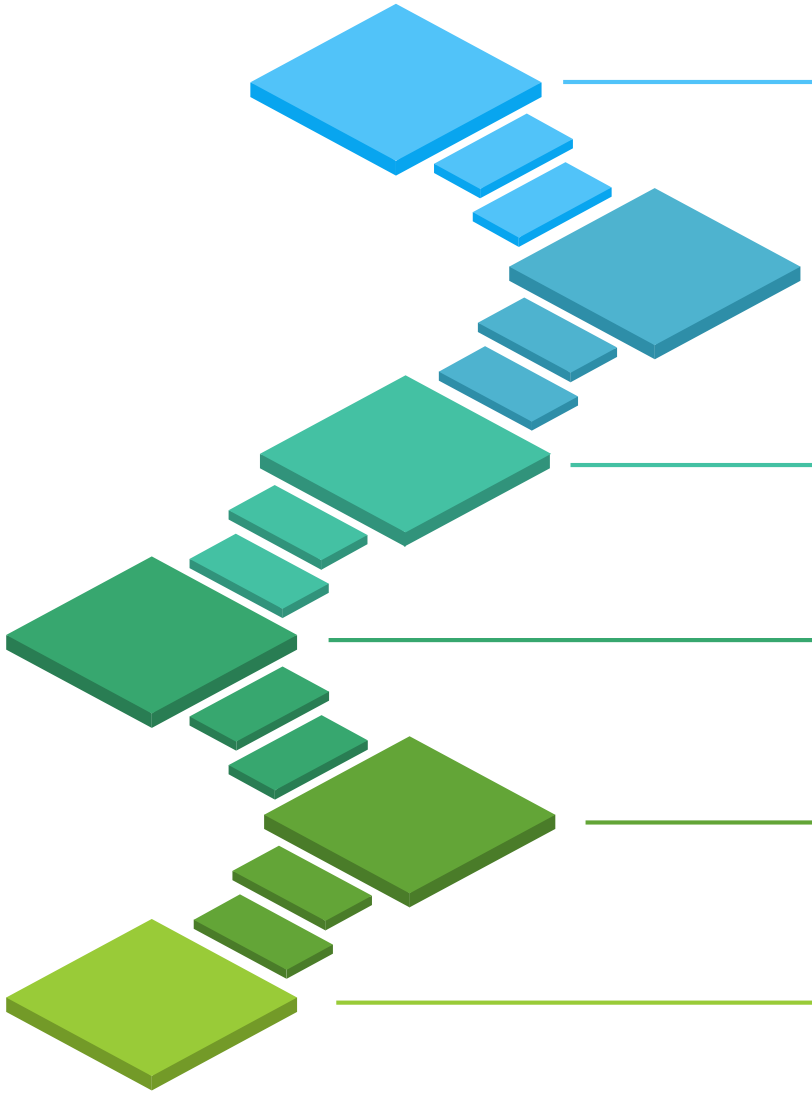
About This Course

In the course you will acquire the skills required in the ongoing work of the Data Analyst in the Python language

Overview



1	Basic Python
2+3	Pandas & NumPy
4+5	Data Transformation
6	SQL in python
7	Data Visualization
8	Intro to ML

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Methodology

- Study topics in class including examples
- Practice in class
- Home practice

Today's Topics

- Python language – brief introduction
- Data types Strings – manipulating strings and regular expressions.
- Basic flow control - conditions and loops
- Data structures - lists, dictionaries, sets and tuples

Introduction to Python

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Introduction to Python

Python is a language developed by Guido van Rossum in 1991

Open source language

- Web application development
- image processing
- artificial intelligence
- automation
- GUI - User interface
- Games

Python became popular among data science community in the last five or so years.

Jupyter Notebook

- An environment where you can write, check and run the code quickly
    - Designed to "tell a story", and conduct research on data
        - Easy to use compared to other Integrated Development Environment
            - Especially popular for data analysts

Cell

- Cell types:
  - Markdown: used to write headlines, comments and images
  - Code: used to write Python code that will be run by the interpreter
- Cell modes:
  - Edit mode
  - Command mode

Keyboard Shortcuts

Shortcut key	Description
Enter	Enter edit mode / line break in cell
Ctrl + Enter	Run cell
Shift + Enter	Run cell, select cell below
Esc → A	Insert cell above
Esc → B	Insert cell below
Esc → D D (press D twice quickly)	Delete selected cell

Operators

Operators	Action	Example
+	Addition	x+y, x+3, y+x+100, 4+5
-	Subtraction	x-5, x-y, y-3-x, 10-3
*	Multiplication	x*y, 30*x, 4*2
/	Division	x/y, x/3, 9/4
**	Exponentiation	x**y, y**2, 2**3
//	Floor Division	x//y, y//9, 10//3
%	Modulo	x%y, x%2, 9%4, 5%2

Basic Data Types

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Variables

A variable is a name granted to an object in a program,

in order to provide the context for it, and for what it represents

A variable definition is performed by the assignment operation:

Variable_name = variable_value

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* Cannot start with number or special characters

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Basic Data Types

Data type	Value	Example
int	Whole number	5, 267, -10, 100000
float	Decimal number	4.98, 378.2, -9.99
str	String	“Talia”, ‘Python Course!!’
bool	Boolean	True, False

* All these data types are immutable objects, can not be modified

Operators

Operators	Action	Example
==	Comparison	x == 2
!=	different	x != 2
>, >=	Greater than / greater than and equal to	x >2, 4<=x<10
<, <=	Small from / Small and equal to	x<2
and	And - both expressions must be true	x>2 and x!=10
or	Or - one of the expressions must be true	x!=3 or y>40
not	Reverse what comes out	not (x<2)
in	Check if the string contains substring	"tho" in "Python course"

Build In Functions

Function is a sequence of instructions to get the desired outcome

In order to invoke function:

function_name(argument1, argument2...)

Print () Function

```
In [12]:  print ("Hello")

          Hello
```

Print text

```
In [13]:  sales_total = 52000

          print (sales_total)

          52000
```

Print variable value

```
In [14]:  sales_total = 52000
          print ("Sales total is: ", sales_total)

          Sales total is:  52000
```

Printing text and variable value

Special Chars For print()

Char	Action	Example
\n	Row brake	print ("go to \nnext line") >> go to next line
\t	Tab	print ("jump to \tnext tab") jump to next tab >>

Data Type String

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Slicing

With slicing we can get substring

string[start : stop : step]

Beginning of the slicing

End position (not include)

Step allow skipping on characters

Slicing

Jump

+3 +3 +3 +3

P	Y	T	H	O	N		C	L	A	S	S	!
0	1	2	3	4	5	6	7	8	9	10	11	12
-13	-12	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1

The access to the characters in the string is according to their index (location)

Count from end to start by using negative index

Method

Method is a function that belongs to a specific object and directly associated with its data type

```
object_name.method_name(parameter1, parameter2...)
```

String Methods

Method	Action
upper()	Converts a string into upper case
lower()	Converts a string into lower case
capitalize()	Converts the first character to upper case
strip()	Remove spaces at the beginning and at the end of the string
replace (old,new,count)	replace occurrences of a substring with a new string
zfill(len)	Fills the string with a specified number of 0 values at the beginning
isalpha()	Returns True if all characters in the string are in the alphabet
isdigit()	Returns True if all characters in the string are digits

* Srting methods

Method Chaining

Method chaining is a sequence of methods being invoked in a method chain

```
object_name.method_1().method_2()
```

Conditions

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Conditions – IF Block

```
if <condition>:
    <what_to_do_if_true>
elif <condition_2>:
    <what_to_do_if_true>
else:
    <what_to_do>
```

Tab

Condition

Extra conditions if needed

Loops

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While

While the condition is true, the loop keep on iterating

```
While  <condition>:
    Tab <do_something>
```

loop

For

Executes the same block of code once for each item in the sequence

```
for <var> in <sequence>:
    <do_something>
```

Tab

loop

Loop Flow Control

break

When we want to immediately exit the block of the loop

continue

When we want to immediately stop the current iteration of the loop and start the next iteration

Data Structures

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Data Structures

Set { }

Set is a collection which is unordered, changeable and unindexed

List [ ]

List is a collection which is ordered, changeable

Dict { : }

Dictionary is a collection which is unordered, changeable and indexed

Tuple ( )

Tuple is a collection which is ordered and unchangeable

List

- List holds ordered collection of items
- Mutable data structure
- Items can repeat

my_list = ['item_1', item_2, …]

List Methods

Method	Action
append('new_value')	Adding an item to the end of the list
pop()	Removes the item at the specified index
remove('value')	Removes the first item with the specified value
sort()	Sort the list
count ('value')	Returns the number of items with the specified value
reverse()	Reverses the order of the list
clear()	Removes all the elements from the list

Tuple

- Tuple holds ordered collection of items
- Immutable data structure
- Items can repeat

my_tuple = ('item_1', item_2, ...)

Tuple Methods

Method	Action
index('value')	Searches the tuple for a specified value and returns the position of where it was found
count ('value')	Returns the number of times a specified value occurs in a tuple

Set

- Set holds unordered collection of items
- Immutable data structure
- Items cannot repeat

my_set = {'item_1', item_2, …}

Set Methods

Method	Action
add('new_value')	Adds an item to the set
remove('value')	Removes the specified item
pop()	Removes a random item from the set
clear()	Removes all the elements from the set

Dictionary - Dict

- A dictionary is a key:value pair
- Mutable data structure
- Key must be unique

```
my_dict = {'key1':'value1',
           'key2':value2, ...}
```

Dictionary Methods

Method	Action
keys()	Returns a list containing the dictionary's keys
values()	Returns a list of all the values in the dictionary
Items ()	Returns a list containing a tuple for each key value pair
clear()	Removes all the items from the dictionary

Casting Functions

Function	Action
Int (x)	Convert a number or string to an integer
Str (x)	Create a new string object from the given object
Float (x)	Convert a string or number to a floating point number, if possible
Bool (x)	Returns True when the argument x is true, False otherwise
List (data_structure_var)	Convert variable from given data structure into list
dict (data_structure_var)	Convert variable from given data structure into dictionary
set (data_structure_var)	Convert variable from given data structure into set
tuple (data_structure_var)	Convert variable from given data structure into tuple

Thank You

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